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Hepatitis Delta and Liver Disease Among People Living With Hepatitis B With or Without HIV Co-Infection in Senegal

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Keywords: hepatitis B | hepatitis D | liver fibrosis | prevalence | Senegal

ABSTRACT

Background and Aims: The prevalence of hepatitis delta virus (HDV) infection among persons living with hepatitis B virus (HBV) and its impact on liver-related complications in West Africa are ill-defined. Wetested a large urban HBV cohort in Senegal for the presence of HDV/HBV co-infection and evaluated its association with liver fibrosis.

Methods: We included persons with positive hepatitis B surface antigen (HBsAg) enrolled in the SEN-B cohort since 2019. Anti-HDV antibodies (HDVAb) were tested using the Anti-HD Diasorin LiaisonXL test, HDV RNA was measured with RT-qPCR and genotyping was determined through sequencing. We used multivariable logistic regression to evaluate the association between HDVAb positivity and liver fibrosis, defined as a liver stiffness measurement > 7.0 kPa.

Results: We analysed 914 individuals with a median age of 32 years (interquartile range [IQR] 26–41), of whom 487 (53.3%) were men and 117 (12.8%) had HIV co-infection. Thirteen participants (1.4%, 95% CI 0.8–2.4) had a positive HDVAb test, of whom 8/13 (61.5%) showed detectable HDV RNA. HDV genotype 5 was found in 75.0% of cases. In multivariable analyses, HDVAb positivity

Abbreviations: ALT, alanine aminotransferase; aOR, adjusted odds ratio; APRI, aspartate aminotransferase to platelet ratio index; AST, aspartate aminotransferase; AUDIT-C, Alcohol Use Disorders Identification Test-Consumption; BMI, body mass index; CI, confidence interval; CRCF, Centre Régional de Recherche et Formation Clinique à la Prise en Charge; CTA, Centre de Traitement Ambulatoire; DNA, deoxyribonucleic acid; g/L, gram per litre; HBeAg, hepatitis B e antigen; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HCC, hepatocellular carcinoma; HCVAb, hepatitis C virus antibody; HDV, hepatitis D virus; HDVAb, hepatitis D virus antibody; HDV-gt, hepatitis D virus genotype; HIVAb, human immunodeficiency virus antibody; IU/L, international units per litre; Kg/m^2, kilograms per square meter; kPa, kilopascal; Log_{10}, logarithm to the base ten; LSM, liver stiffness measurement; Neg, negative; Pos, positive; pwHBV, persons living with hepatitis B; qHBsAg, quantitative hepatitis B surface antigen; RNA, ribonucleic acid; SEN-B, Senegalese hepatitis B cohort study; SMIT, Service de Maladies Infectieuses et Tropicales; TDF, tenofovir disoproxil fumarate; TE, transient elastography.

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(aOR 11.7, 95% CI 3.1–45.7), male sex (aOR 5.4, 95% CI 3.1–10.3), ALT > 40 IU/L (aOR 4.4, 95% CI 2.4–8.2) and HBeAg positivity (aOR 4.6, 95% CI 1.8–11.9) were independently associated with liver fibrosis.

Conclusion: The prevalence of HDV infection was low in persons living with HBV in Dakar, but those affected had a very high risk of presenting with liver cirrhosis. Efforts to improve HDV screening and management are urgently needed in Senegal.

1 | Introduction

Approximately one-fourth of the 64 million persons living with chronic hepatitis B virus (HBV) infection in Africa are also infected with hepatitis delta virus (HDV), which relies on the presence of hepatitis B virus surface antigen (HBsAg) to infect liver cells for entry into hepatocytes [1, 2]. HDV infection is one of the most severe forms of viral hepatitis, being associated with a rapid progression of liver fibrosis and an increased risk of hepatocellular carcinoma (HCC) [3, 4]. In addition to older age, liver cirrhosis and co-infection with HIV, ongoing HDV replication has been identified as an independent predictor of progression to cirrhosis and HCC among persons with HBV [4–6].

The prevalence of HDV varies across different regions in Africa, being highest in Central Africa and lowest in Eastern and Southern Africa [2, 7]. High-prevalence spots were reported in the Democratic Republic of the Congo (26%), Gabon (25%–50%), Guinea-Bissau (23.9%) and Cameroon (14%–35%) [3, 7, 8]. In a nationwide study conducted in Cameroon, the overall prevalence of HDV antibodies (HDVAb) was 11%, with estimates ranging from 1% in the North to 54% in the South [9]. A higher risk of HDV infection has been documented in specific groups of the population such as people living with HIV and people who inject drugs [7, 10, 11]. Despite the high prevalence of chronic HBV infection in Senegal, few studies focused on HBV/HDV co-infection [12]: Small studies have reported a prevalence ranging from 3% to 20% among blood donors and people with HIV/HBV co-infection [13–15].

While the distribution of HBV genotypes in African regions has been extensively characterised, there are limited data on the circulating HDV genotypes. HDV genotype 1 has been found worldwide, including in Africa, whereas HDV genotypes 5 to 8 appear to be exclusive to Africa [2, 7]. Genotypes 1, 2 and 5 have been associated with severe liver disease in several cohort studies from Europe and Asia. However, results may vary by population and context, as reported in a London-based cohort study, where genotype 5 was instead associated with more favourable disease outcomes [5, 16, 17]. The impact of HDV genotypes on the natural progression of HDV-related liver disease is not well defined in Africa. The recent development of novel antiviral therapies for HDV infection has reinforced the need to understand the HDV epidemiological landscape and genotype distribution, particularly in Africa [18].

We aimed to determine the prevalence of chronic HDV co-infection among people with HBV (pwHBV) in urban Senegal, to characterise HDV genotypes and evaluate the association between HDV infection and liver fibrosis.

2 | Material and Methods

2.1 | Study Design and Participants

We performed a cross-sectional analysis of adults with laboratory-confirmed HBsAg-positive tests with or without HIV co-infection, enrolled in the Senegalese Hepatitis B Cohort (SEN-B) from September 2019 to December 2023. SEN-B cohort is a prospective, population-based cohort of people living with hepatitis B in urban Senegal, which enrolled consecutive persons living with HBV presenting at two participating clinics [19]. All participants were enrolled at two main referral HIV clinics in Senegal: the Service des Maladies Infectieuses et Tropicales (SMIT) and the Centre de Traitement Ambulatoire (CTA), both located at Fann University Hospital in Dakar.

2.2 | Study Procedures and Data Collection

At study enrolment, we collected extensive data on sociodemographic, behavioural, HBV transmission and clinical risk factors. Serum, plasma and dried blood spots (DBS) were collected and cryopreserved at −80°C for further serological and molecular analyses. Individual data were collected and anonymised using standardised case report forms and entered into a centralised electronic database (ARPEGE). Socio-demographic variables included age, sex, level of education and country of origin. Alcohol consumption was assessed using the Alcohol Use Disorders Identification Test-Consumption (AUDIT-C) [20]. Hazardous drinking was defined as an AUDIT-C score > 3 for women and > 4 for men. We defined a family history of HCC as having a first-degree relative with a history of liver cancer, according to the participants' reports. The use of traditional medicine was defined as any oral ingestion or topical application of plant-based preparations (leaves, roots or tree bark) in the past 6 months before study inclusion.

We tested for the presence of HBsAg using a lateral-flow assay test (NovaTest, Atlaslink-inc ISO 13485). The Elecsys HBsAg II and Elecsys HBeAg assays were used for HBsAg confirmation and detection of HBeAg using the Roche e411 cobas (2011–08 V.10) platform. The quantitative hepatitis B surface antigen (qHBsAg) was measured using the e411 cobas HBsAg II assay (Roche Diagnostics, Rotkreuz, Switzerland), a commercial chemiluminescent microparticle immunoassay with a sensitivity of ≤ 0.05 IU/mL. HBV DNA levels were measured using the cobas/TaqMan (Roche Diagnostic Systems, Meylan, France) with a lower limit of quantification (LLOQ) of 20 IU/mL. HIV and hepatitis C virus (HCV) were tested using commercial lateral-flow assays (Determine HIV, Bioline HCV; Abbot, Chicago, USA). Platelet levels were measured using an automated haematology

Summary

- Among 914 persons living with hepatitis B in Dakar, Senegal, 1.4% had hepatitis delta virus (HDV) infection.
- Those with HDV infection had a much higher risk of severe liver disease, emphasising the need for improved HDV screening and management in Senegal.

analyser (BC-3000plus, Minday Bio-Medical Electronics Co. Ltd. Shenzhen, China). The Indiko Clinical Chemistry Analyzer (Thermo Scientific, Massachusetts, US) was used for alanine aminotransferase (ALT) and aspartate aminotransferase (AST). The upper limit of normal for ALT and AST was defined as 40 IU/L [21]. AST to platelet ratio index (APRI) score was calculated using the following formula: AST Level (IU/L) ×100/AST (Upper Limit of Normal [40 IU/L])/Platelet Count (10^9/L). An APRI score > 0.5 was used to define significant liver fibrosis [22].

Liver fibrosis was assessed using transient elastography (TE) by a single trained operator according to the instructions provided by the manufacturer (Fibroscan, Echosens, France). We used an XL probe if BMI was > 25 kg/m^2. Significant liver fibrosis was defined as a liver stiffness measurement (LSM) > 7.0 kPa, and cirrhosis as an LSM > 12.5 kPa as per WHO thresholds [22]. An examination was deemed successful if at least 10 valid LSMs were obtained and considered reliable if the interquartile range divided by the median (IQR/M) was < 0.3 [23].

2.3 | Hepatitis D Laboratory Tests

All HDV tests were performed using dried blood spots (DBS, Lasec, Cape Town, South Africa) cryopreserved at −80°C at the Laboratory of the Centre Régional de Recherche et Formation Clinique à la Prise en Charge de Fann (CRCF), Fann University Hospital, Dakar, Senegal. The DBS were prepared from samples at or closest to enrolment using whole blood, serum or plasma if available. For each participant, two DBS spots were eluted in 1 mL phosphate-buffered saline (PBS) buffer containing 0.01% NP40 (NP-40 Surfact-Amps Detergent Solution, Massachusetts, US) for 1 h shaking. Total HDVAb was initially measured using the Anti-HD Liaison XL test (DiaSorin, Belgium). Indeterminate or weak positive results (Ratio values between 0.5–1.5) were confirmed by a second anti-HDV ELISA assay (Dia.Pro Diagnostic Bioprobes Srl, Milan, Italy). The quantification of HDV RNA was performed using the RT-qPCR, EurobioPlex EBX 071 kit (Eurobio-France), after RNA extraction with the Abbott m2000sp (Abbot Laboratories, North Chicago, US) device from 500 μL of the sample. The recently validated HDV RNA lower limit of quantification (LLOQ) of 1.7 log IU/mL was used [24]. For all samples with an HDV RNA > 1.7 log IU/mL, a sensitive qualitative RT-PCR named *'R0'* was performed, and the obtained amplicon was sequenced bidirectionally using the Sanger method (BigDye Terminator v3.1-Thermofisher Scientific) for genotype characterisation.

2.4 | Statistical Analyses

HDVAb-positivity prevalence was calculated with a 95% confidence interval (CI). Participants' characteristics were compared between individuals with positive and negative HDVAb using Fisher's test for categorical variables and Wilcoxon rank-sum test for continuous variables. We used multivariable logistic regression to evaluate the association between HDV infection and the following outcomes: (i) liver fibrosis (LSM > 7.0 kPa) and (ii) liver cirrhosis (LSM > 12.5 kPa). We adjusted the multivariable models for the following potential confounders: age, sex, country of origin, body mass index, HIV status, HBeAg, qHBsAg and ALT. *p*-values below 0.05 were considered statistically significant. All statistical analyses were performed using R Statistical Software version 4.4.0 [25]. The population attributable fractions (PAF) for significant liver fibrosis/cirrhosis due to HDVAb positivity was calculated using Levin's formula [26]: $PAF = p \times (RR-1)/p \times (RR-1) + 1$ where 'p' represents the prevalence of HDVAb positivity in the study population not affected by liver fibrosis/cirrhosis. The relative risk (RR) corresponding to the risk of liver fibrosis/cirrhosis associated with HDVAb positivity was approximated through the aOR obtained from the multivariable regression models. The main packages we used included: Tidyverse (version 2.0.0) for data manipulation (dplyr), visualisation and plots (ggplot2) and sjPlot (version 2.8.16) for regression models [27, 28].

3 | Results

3.1 | Study Population Characteristics

Of 914 SEN-B participants, 487 (53.3%) were men, 117 (12.8%) were living with HIV and the median age was 32 years (interquartile range [IQR]: 26–41). At enrolment, all pwHIV-HBV were on a TDF-containing antiretroviral treatment (ART) for a median duration of 63.8 months (IQR: 32.2–95.2). Among 797 pwHBV, 50 (6.3%) were on TDF, with a median treatment duration of 9.7 months (IQR: 4.2–26.6). Among 747 TDF-naïve pwHBV, 114 (15.3%) had undetectable HBV DNA.

3.2 | Hepatitis D Infection Prevalence and Determinants

Overall, 13 of 914 participants had a positive HDVAb test, resulting in a prevalence of 1.4% (95% CI 0.8–2.4). Of 13 participants with confirmed HDV infection, three exhibited weak positivity in the Anti-HDV Liaison XL test, which was subsequently confirmed by the anti-HDV ELISA assay. The prevalence of HDVAb positivity was 1.3% (10/797; 95% CI 0.68–2.29) among pwHBV and 2.6% (3/117; 95% CI 0.87–7.27) among pwHBV-HIV. Compared to individuals with negative HDVAb, those with positive HDVAb were more likely to have a non-Senegalese African origin (23.1% vs. 2.4%; $p = 0.004$), a family history of HCC (30.8% vs. 10.9%; $p = 0.047$), elevated AST (30.8% vs. 6.5%, $p = 0.01$), lower median platelet count (213 10^9/ vs. 282 10^9/L, $p = 0.01$) and APRI-score > 0.5 (38.5% vs. 4.4%, $p < 0.001$; Table 1). Eight of 13 participants with positive HDVAb (61.5%) showed active HDV RNA replication, of whom four (30.8%) had HDV RNA levels ≥ 3 $\log_{10}$ IU/mL. Six of the eight participants with detectable HDV RNA were TDF-naïve, of whom

TABLE 1 | Characteristics of the study participants by HDVAb status.

Characteristics	HDVAb-negative $N=901$	HDVAb-positive $N=13$	p
Age (years)	32 (26–41)	36 (27–40)	0.50
Male sex	479 (53.2)	8 (61.5)	0.50
Secondary education level or higher	246 (27.3)	6 (46.2)	0.20
Non-Senegalese origin[a]	22 (2.4)	3 (23.1)	0.004
Family history of HCC	98 (10.9)	4 (30.8)	0.047
History of tattoos or scarification	73 (8.1)	1 (7.7)	0.99
History of surgery	124 (13.8)	4 (30.8)	0.10
Circumcision or excision	497 (55.2)	9 (69.2)	0.30
History of blood transfusion	26 (2.9)	1 (7.7)	0.30
Smoking	110 (12.2)	2 (15.4)	0.70
Hazardous drinking	18 (2.0)	1 (7.7)	0.20
Use of traditional medicine	297 (33.1)	4 (30.8)	0.99
BMI (kg/m^2)	22 (19.6–25.4)	20.9 (17.6–25.0)	0.40
BMI ≥ 30.0	59 (6.6)	2 (15.4)	0.20
HCVAb positivity	1 (0.1)	0 (0.0)	0.99
HIVAb positivity	114 (12.7)	3 (23.1)	0.20
AST (IU/L)	21 (16–26)	31 (22–49)	0.01
> 40.0	59 (6.5)	4 (30.8)	0.01
ALT (IU/L)	18 (13–24)	24 (15–32)	0.14
> 40.0	72 (8.0)	2 (15.4)	0.30
Platelets (10^9/L)	284 (241–337)	213 (165–276)	0.01
HBeAg positivity	24 (2.7)	1 (7.7)	0.30
qHBsAg > 1000 (IU/mL)	688 (76.4)	10 (76.9)	0.99
APRI score > 0.5	40 (4.4)	5 (38.5)	< 0.001
On TDF at enrolment	151 (16.76)	7 (53.85)	0.003

Note: Categorical variables are presented as No. (%) and quantitative ones as median (IQR).
Abbreviations: ALT: alanine aminotransferase, APRI: aspartate aminotransferase to platelet ratio index, AST: aspartate aminotransferase, BMI: body mass index, DNA: deoxyribonucleic acid, L: litre, HBeAg: hepatitis B e antigen, HBV: hepatitis B virus, HCC: hepatocellular carcinoma, HCVAb: hepatitis C virus antibody, HDVAb: hepatitis D virus antibody, HIVAb: human immunodeficiency virus antibody, IU/L: international units per litre, IU/mL: international units per millilitre, Kg/m^2: kilograms per square meter, qHBsAg: quantitative hepatitis B surface antigen, TDF: tenofovir disoproxil fumarate.
[a]Ivory Coast ($n=4$); Cameroon ($n=1$); Democratic Republic of the Congo ($n=1$); Djibouti ($n=1$); Gambia ($n=1$); Guinea ($n=4$); Guinea-Bissau ($n=5$); Togo ($n=1$); Mauritania ($n=3$).

4 (66.7%) had undetectable HBV DNA (Figure 1). We sequenced HDV in eight participants with detectable HDV RNA: Six of them (75.0%) had HDV genotype 5 and the two remaining had genotype 1 and 7 (Figure 1). The individuals with genotype 1, and three of those with genotype 5 reported a family history of HCC. Five (62.5%) of eight participants with HDV replication had an LSM > 12.5 kPa: those with genotype 1 and genotype 7, along with three of those with genotype 5 (Table S1).

3.3 | Predictors of Liver Fibrosis and Cirrhosis

The proportion of participants with LSM > 12.5 kPa was higher among those with positive HDVAb compared to those with negative HDVAb (46.2% vs. 2.4%, $p<0.001$). There was no significant difference in the proportion of participants with ALT > 40 IU/L (15.4% vs. 8.0%, $p=0.30$) or qHBsAg > 1,000 IU/mL (76.9% vs. 76.5%, $p=0.99$) between individual with positive and negative HDVAb (Figure 2). Participants with detectable HDV RNA had higher median ALT (31 IU/L [IQR: 26.4–36.1] vs. 15 IU/L [IQR: 12.2–16.7], $p=0.03$) and median LSM (16 kPa [IQR: 7.5–21.3] vs. 6 kPa [IQR: 4.6–6.8], $p=0.03$) than those with undetectable HDV RNA (Figure 3). In multivariable analyses, HDVAb positivity was strongly associated with significant liver fibrosis (adjusted odds ratio [aOR] 11.7, 95% CI 3.1–45.7) and liver cirrhosis (48.9, 11.2–228.0, Figure 4, Table S2). Levin's PAF for liver fibrosis attributed to HDVAb positivity was 7.3% for fibrosis and 27.5% for cirrhosis. Other factors associated with liver fibrosis

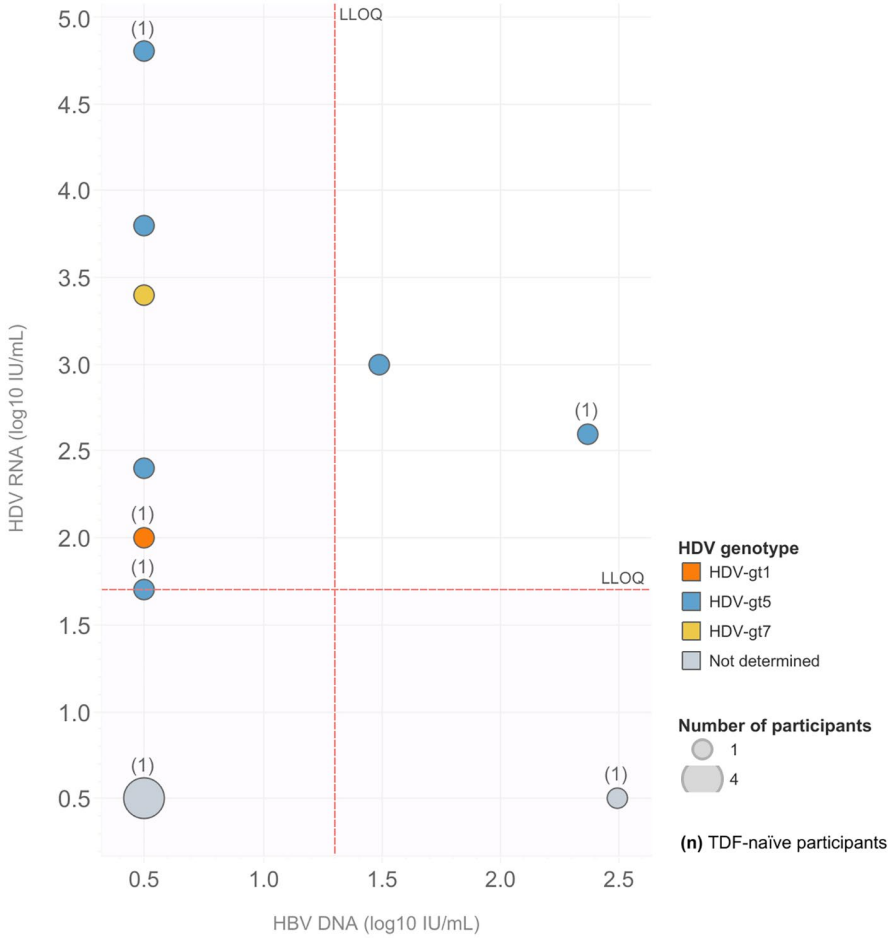

FIGURE 1 | HDV RNA and HBV DNA levels by HDV genotypes among individuals HDVAb positive. HBsAg: hepatitis B surface antigen, HDV: hepatitis D Virus, HDVAb: hepatitis D virus antibody, HDV-Gt: HDV genotype, IU/mL: international units per millilitre, LLOD: lower limit of detection, LLOQ: lower limit of quantification, RN A: ribonucleic acid. All HDV RNA and HBV DNA values below the LLOQ were arbitrarily assigned a value of $0.5 \log_{10}$ IU/mL.

were male sex (aOR 5.4, 95% CI 3.1–10.3), ALT >40 IU/L (aOR 4.4, 95% CI 2.4–8.2) and HBeAg positivity (aOR 4.6, 95% CI 1.8–11.9). When defining HDV infection by HDV RNA positivity, we observed strong associations with liver fibrosis (aOR 23.0, 95% CI 4.1–194) and cirrhosis (aOR 56.9, 95% CI 10.0–405; Table S3).

4 | Discussion

In our large cohort of persons living with HBV in urban Senegal, we found a very low prevalence of HDV infection, independent of the HIV status. HDV RNA replication was observed in more than 60% of participants with HDVAb positivity, and genotype 5 was confirmed in the majority of participants with active HDV replication. HDV infection was a strong risk factor for liver cirrhosis in our cohort: 62.5% of participants with HDV replication had liver cirrhosis at the time of diagnosis. Our results show that despite HDV infection being rare in our setting, its diagnosis is crucial given the high risk of severe liver disease in this population.

The prevalence of HDV infection in our study was 1.4%, much lower than the 20% reported in a study conducted among 90 individuals with positive HBsAg in Senegal in 1992 [13]. The highest prevalence in that study may have been overestimated due to the suboptimal sensitivity and specificity of the assay used and selection bias [29]. Furthermore, the introduction of the infant HBV vaccination in Senegal in 2004 had a significant impact on the epidemiology of HBV, which may have contributed to a reduction in HDV transmission [30]. Similar findings to ours were recently reported in population-based studies from West Africa, which showed an HDV prevalence of 1.4% in the Gambia and 1.1% in Burkina Faso [31, 32]. This variability in HDV seroprevalence reflects the geographical heterogeneity of HDV distribution across Africa, and sometimes even within the same country [2, 7]. For instance, the highest HDV seroprevalence seen in Gabon (>66.0%) was correlated with the presence of specific ethnic groups in the northeastern provinces [33]. A comparable intracountry heterogeneity was also recently described in Cameroon and Nigeria [9, 34].

In our study, participants with positive HDVAb were 12 times more likely to present with significant fibrosis compared to those with negative HDVAb. This association was twice as strong when considering RNA positivity as the parameter defining hepatitis delta. In addition, more than half of the participants with HDV

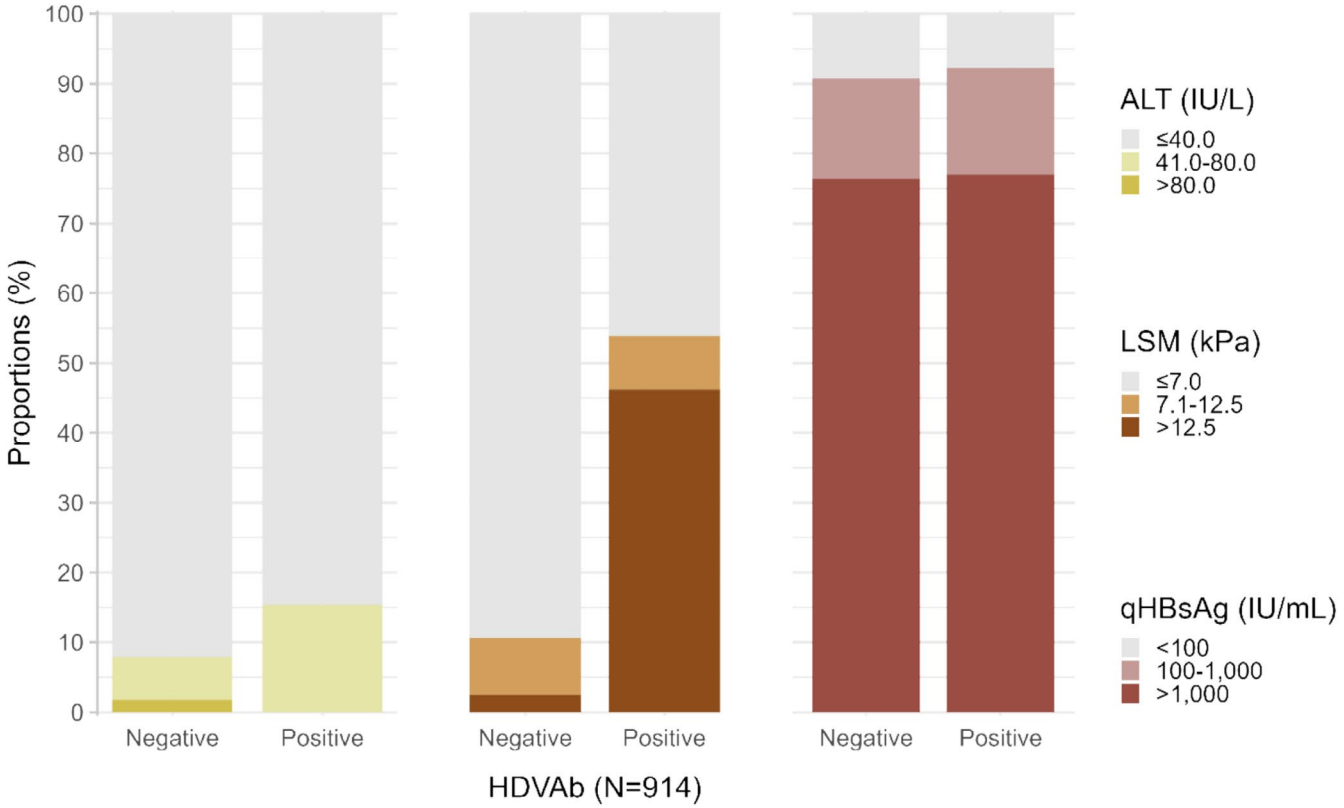

FIGURE 2 | ALT, LSM and qHBsAg levels by HDVAb status. ALT: alanine aminotransferase, HDVAb: hepatitis D virus antibody, IU/L: international units per litre, kPa: kilopascal, LSM: liver stiffness measurements, qHBsAg: quantitative hepatitis B surface antigen.

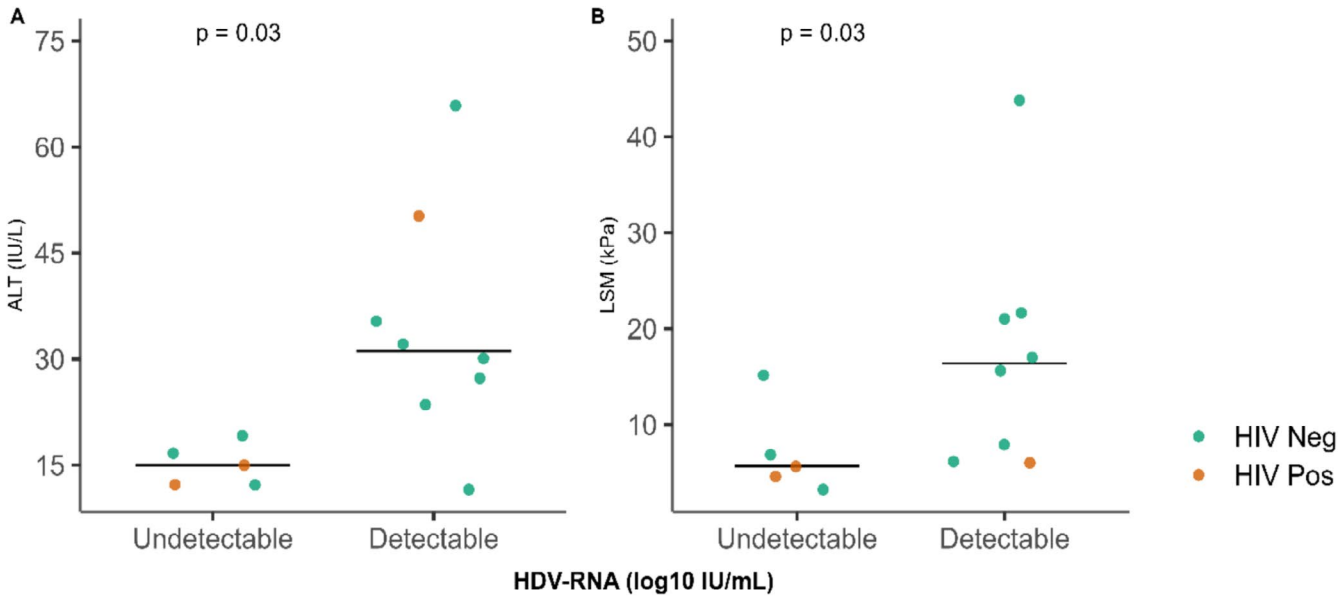

FIGURE 3 | ALT (A) and LSM (B) by HDV RNA detectability status ($N = 13$). Each dot represents individual data. ALT: alanine aminotransferase, HDV: hepatitis D virus, HIV: human immunodeficiency virus, IU/L: international units per litre, kPa: kilopascal, $\log_{10}$: logarithm to base 10, LSM: liver stiffness measurements, Neg: negative, Pos: positive, RNA: ribonucleic acid.

infection had liver fibrosis or cirrhosis, and among eight persons with HDV replication, five had liver cirrhosis. In a French cohort study including over 1100 individuals with HDV infection, with the majority coming from sub-Saharan African countries, 36.6% had significant or severe fibrosis at enrolment [5]. Given the high burden of advanced liver disease in the presence of active HDV replication, the WHO has recently included reflex testing for HDV infection in its latest HBV guidelines [22].

Eight of the 13 participants (61.5%) with positive HDVAb had detectable RNA in the plasma, of whom four (50.0%) had high-level viral loads. This finding aligns with several studies from different

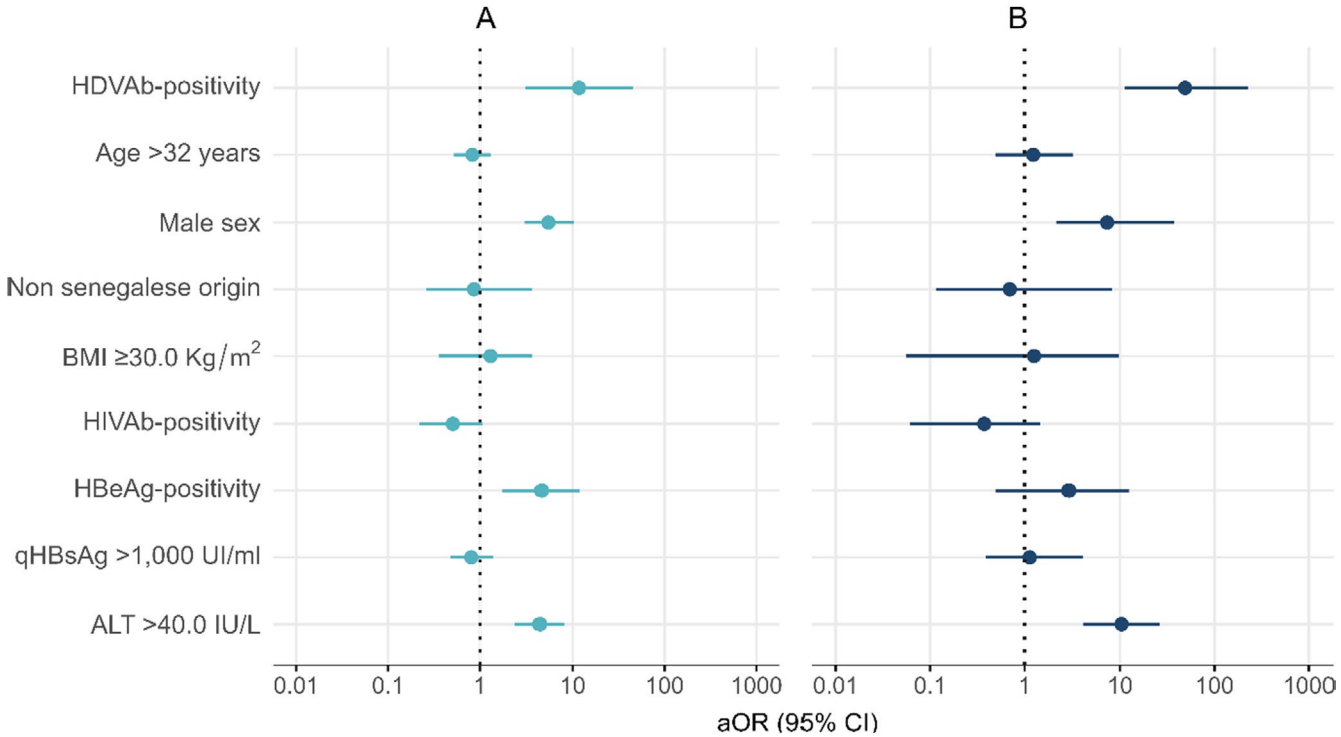

FIGURE 4 | Adjusted association of HDVAb positivity with (A) liver fibrosis (LSM > 7.0 kPa) and (B) cirrhosis (LSM > 11.0 kPa). ALT: alanine aminotransferase, aOR: adjusted odds ratio, BMI: body mass index, CI: confidence interval, HBeAg, hepatitis B envelope antigen, HDVAb: hepatitis D virus antibody, HIV: human immunodeficiency virus, Kg/m^2: kilograms per square metre, LSM, liver stiffness measurement.

settings indicating that two-thirds of individuals tested positive for HDVAb have a replicating HDV infection at the time of diagnosis [35, 36]. In our study, two-thirds of TDF-naïve (66.7%) participants with detectable HDV RNA had undetectable HBV DNA, in line with our general understanding that HDV replication often dominates over HBV replication. These findings further support the need for reflex testing for HDV, as individuals with unrecognised HDV infection may seem ineligible for HBV treatment due to low HBV DNA. Our participants with detectable HDV RNA had higher median ALT levels and LSM compared to those with undetectable HDV RNA. Similarly, in a multicentre study conducted across four hospitals in Spain, individuals with detectable HDV RNA were more likely to present with elevated ALT levels and cirrhosis compared to those with undetectable HDV RNA [36].

HDV genotypes identified in our study (HDV-gt5, HDV-gt1, HDV-gt7) belong to clades traditionally reported in Africa [2, 7]. HDV-gt5 was the most prevalent genotype, in line with global and regional studies indicating its predominance in West Africa [2, 37]. The only participant with HDV-gt7, was from Cameroon, indicating a probable infection prior to his arrival in Senegal. Whether genotypes are responsible for different degrees of severity of liver disease, has not yet been explored in Africa. Analyses from a French cohort showed that European individuals with HDV-gt1 and African individuals with HDV-gt5 had a higher likelihood of developing liver cirrhosis than those with other genotypes [5]. Large multicountry studies of individuals living with HBV across Africa are needed to address this question.

Our study is one of the largest in Africa with systematic screening for HDV infection using validated serological assays, and molecular diagnostic methods ensuring reliable results. Moreover, we provide the first indications of the HDV genotypes circulating in Senegal. The low proportion of participants with active HDV replication limited our ability to assess the impact of specific HDV RNA levels and genotypes on liver-related complications. As in most studies relying on noninvasive methods to assess liver fibrosis, there may be some uncertainty about the accuracy of liver fibrosis diagnosis, given that transient elastography has not been adequately validated for individuals with chronic HDV infection. However, it has been shown that transient elastography performs better than other noninvasive tests in the detection of liver cirrhosis in persons with HDV infection [38].

5 | Conclusion

The prevalence of HDV infection is low in urban Senegal. The strong association between HDVAb positivity and advanced liver disease emphasises the importance of systematic HDV testing for all people living with HBV in Senegal. Screening for HDV replication is crucial, as people with active HDV infection are at high risk of complications. While widespread HBV vaccination is essential for preventing HDV infection in Africa, broader access to HDV therapies is crucial to reduce adverse health outcomes.

Author Contributions

B.S.W., A.R.M. and G.W. contributed to the conceptualisation, methodology and wrote the first draft of the manuscript. B.S.W. and A.R.M. conducted statistical analyses. B.S.W., A.R.M., O.N., A.S.B., H.M.A. and M.S.P. contributed to data collection. F.L.G., C.B., A.R.M. and G.W. made

critical revisions to the manuscript. All authors read and approved the final manuscript.

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Ethics Statement

The study was conducted in accordance with the Declaration of Helsinki and approved by the Senegalese National Health Research Ethics Committee (CNERS) at the Health and Social Action Ministry of Senegal (0061/MSAS/DPRS/CNERS). All participants signed an informed consent.

Conflicts of Interest

G.W. received research grants from Roche Diagnostics and Gilead Sciences and served on advisory boards for ViiV, Gilead Sciences and MSD. All other authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses or interpretation of data; in the writing of the manuscript, or in the decision to publish the results.

Data Availability Statement

Aggregate data from the SEN-B cohort can be obtained by following a formal request procedure. Researchers interested in accessing the data are advised to contact the corresponding author (A.R.) or the principal investigators (M.S., N.F.G. and G.W.). The sharing of data will be done in compliance with relevant data protection regulations and ethical considerations.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.